

SYSTEMS ENGINEERING

Systems Analysis & Design

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2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



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What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields, such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
- **Systems Engineering** is a **holistic** approach to engineering that focuses on how to **design** and manage **complex systems** over their **lifecycle**.



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Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for **designing, implementing, and maintaining complex systems**.
- A **Systems Engineer** must have a broad understanding of engineering, mathematics, science, and technology.
- A **Systems Engineer** must be able to analyze and solve problems in a systematic and holistic way.



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- **Systems Engineering** can be applied to **understand, analyze, and improve human activities**.
- **Systems Engineering** can help us **design and manage complex systems** such as **organizations, cities, economies, and societies**.



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Organizations as Systems

- **Organizations** can be **viewed** as: a **rational system**, a **natural system**, or an **open system**.
- A **rational system** is a **formal organization** that is **designed** to **achieve specific goals** as a **machine**.
- A **natural system** is an **informal organization** that is **emergent** and **adaptive** based on **human interactions**.
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Synergy

- **Synergy** is a simple but powerful concept: **the whole is greater than the sum of its parts**.
- It means the **interactions** could **boost the capabilities** of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you could define a **computational machine**.

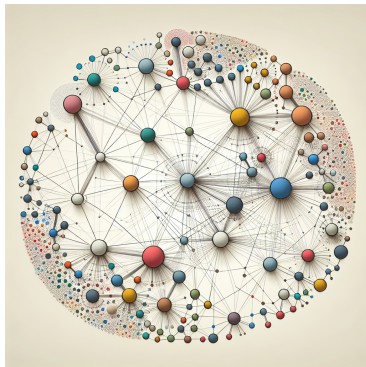


Figure: Prompt: Define a draw of clusters in social networks.



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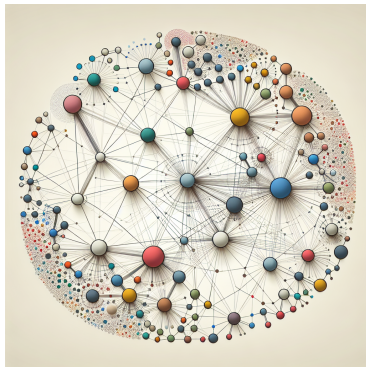
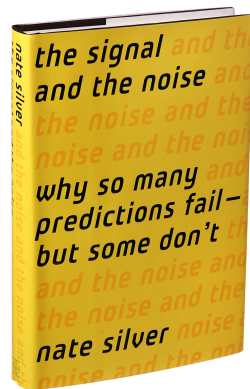


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Synergy: Money Ball



Talking with Machines!

- **Alan Turing** proposed a hundred years ago an **Universal Machine**, capable of take any **algorithm** defined as a **state machine**, and process it in a **binary language**.
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Programming Languages

- **Programming Languages** with more **capabilities**, easier **comprehension** had been created. Also, more people start to **code** into specific **domain programming languages**.

Demo time!

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Applications of Systems Engineering

- **Systems Engineering** can be applied to **understand**, **analyze**, and **improve complex systems**.
- **Systems Engineering** can be applied to **design** and **manage complex systems** such as **organizations**, **cities**, **economies**, and **societies**.
- **Systems Engineering** can be applied to **design** and **manage complex systems** such as **software**, **hardware**, and **networks**.



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Cibernetics

- **Cibernetics** is the **study** of **systems**, **control**, and **communication** in *animals, machines, and organizations*.
- Cibernetics is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.
- Cibernetics is the foundation of systems engineering and information technology.



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Technology

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Technology: AI

- **Artificial Intelligence** is a *field* of computer science that *studies* how to **design** and **implement intelligent agents**.
- Artificial Intelligence is the key to *designing* and *managing* **complex systems** such as smart-organizations, smart-cities, smart-economies, and smart-societies.

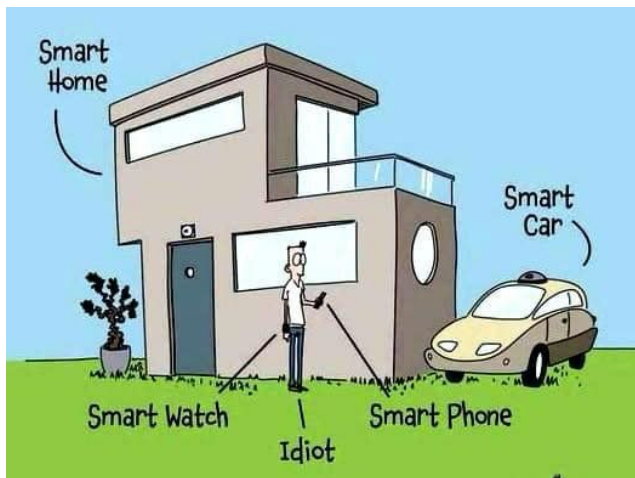


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Artificial Intelligence **not** as a System



Cibernetics & Technology in Systems Context

- **Cibernetics** and **technology** are the **foundation** of **systems engineering** and **information technology**.
- **Cibernetics** and **technology** are the **foundation** of **artificial intelligence** and **smart systems**.



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Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to **make decisions** and **solve problems**.
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Teams as a System

- **Teams** are a **system** where collaboration (synergy) and communication are **key** to **success**.
- Defining and automating the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.



Soft Skills

- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and harmoniously with other **people**.
- Typical Soft Skills:
 - **Communication** skills (verbal and written).
 - **Teamwork** and collaboration.
 - **Problem-solving** and critical thinking.
 - **Adaptability** and flexibility.
 - **Time management** and organization.
 - **Leadership** and management.
 - **Emotional intelligence**.
 - **Creativity** and innovation.
 - **Conflict resolution**.
 - **Networking** and relationship building.
 - **Customer service** and client management.



Computer Analyst

● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
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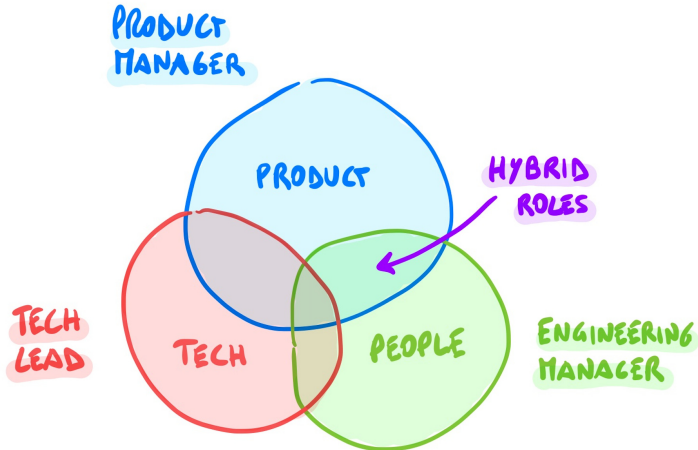
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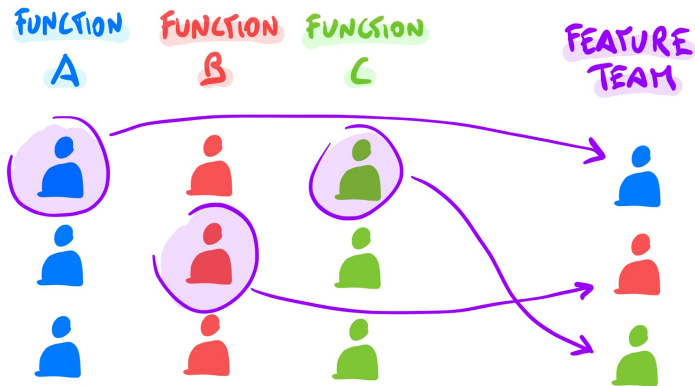
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Tech Company Typical Structure



Feature Teams



What is to be a leader? I

- **Leading** a team is **not a role**. It is a decision, you could be a **leader** anytime and anywhere.
- **Teamwork culture** is pretty **important**. It creates habits, open communication, safety spaces for inclusion.
- **Psychological safety** is a **key** point to have an **effective** team. You could develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. Hierarchy exists by **status** and **power**.
- In a **hierarchy experts lead** to make **better decisions**. However, anyone must be **careful** to not **leave people behind**.



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What is to be a leader? II

- With **crystal-clear communications** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just complete tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish — just **fix and learn**.
- Some believe you're born a leader, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are vital.
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- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



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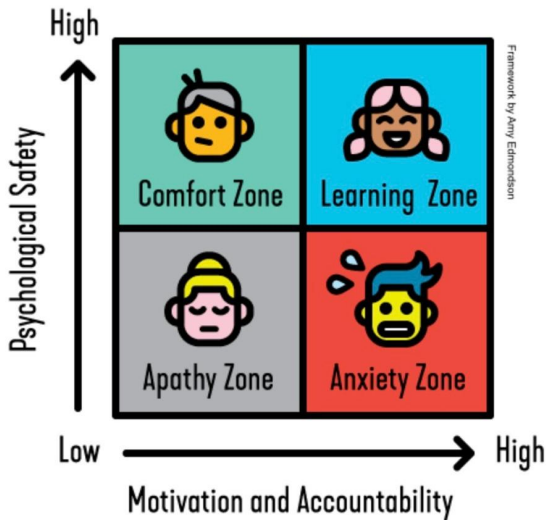


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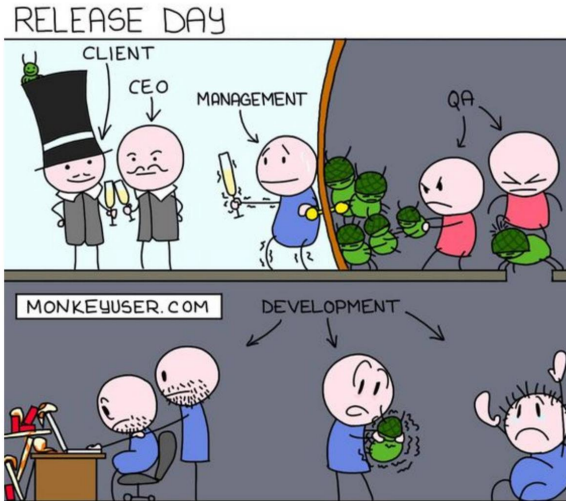
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Working Zones



Real World!



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Thanks!

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Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

